

P S Mithul Sourav

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EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal

Aug 2023 – Jun 2027

Bachelor of Technology in Mechanical Engineering

CGPA: 7.37/10

Selected coursework: Mechatronics; Control Systems; Instrumentation and Measurement; Finite Element Methods; Engineering Mathematics; Probability and Statistics.

RESEARCH INTERESTS

Compliant and variable-stiffness robotic mechanisms; dynamics and control of mechatronic systems; structural dynamics and vibration; computational modeling and simulation; autonomous robotic systems.

RESEARCH AND TECHNICAL PROJECTS

Design and Development of a Tensegrity Based Variable Stiffness Joint

May 2026 – Present

Ongoing project | MATLAB

- Developing an element-level force model of a tensegrity knee joint to study load paths and sensitivity of joint response to geometry and loading.
- Defined three engineering hypotheses and a structured simulation plan; comparing controlled cases in MATLAB to identify parameters for future prototype testing and design refinement.

Prediction of Reaction Wheel Induced Microvibrations Using FEM and Machine Learning

Aug 2026 – Present

Ongoing course project | ANSYS Mechanical, Python

- Building modal and harmonic finite-element models of a satellite panel under reaction-wheel excitation to estimate vibration at a camera mounting interface across the wheel operating-speed range and identify resonant regions.
- Generating a parametric FEM dataset and evaluating regression-based predictions on held-out FEM cases, with specific attention to model behavior near resonance.

Uncertainty-Aware Navigation for an Indoor Mobile Robot

Jun 2026 – Jul 2026

Completed independent simulation project | Python

- Developed a differential-drive simulator with A* planning, heading control, encoder/IMU noise, yaw-bias drift, wheel-slip events, and extended Kalman filter localization.
- Evaluated confidence-aware speed and recovery logic through Monte Carlo trials measuring success, collisions, localization error, recovery events, and time to goal.

NeoLeg Passive Spring-Assisted Knee Mechanism

Dec 2025 – Mar 2026

Completed team project | SolidWorks, ANSYS

- Co-designed a passive spring-assist knee mechanism and contributed to the complete SolidWorks knee-joint assembly and ANSYS analysis.
- Verified interference-free modeled motion through approximately 75–120 degrees of flexion and supported evaluation of spring and attachment geometry.

Off-Road Leaf-Collection Robot

Jul 2026 – Present

NITK IDEA Factory | Mechanical design, mechatronics

- Developing system architecture for locomotion, leaf pickup, debris transfer, storage, and terrain-following subsystems for operation on uneven outdoor terrain.
- Benchmarked existing products, compared mechanism concepts against integration constraints, built a bottom-up BOM, and defined test metrics for traction, pickup efficiency, soil rejection, and endurance.

ENGINEERING EXPERIENCE

Vayu Aerospace Pvt. Ltd.

Jun 2026 – Jul 2026

Product Intern

Bengaluru, India

- Built 3D SolidWorks models of UAV avionics hardware from measured dimensions, mounting points, and connector locations; integrated them into the existing assembly to check fit, clearances, cable routing, connector access, and serviceability.
- Supported UAV hardware bring-up by checking electrical connections and the power path around the flight controller before calibration and motor testing; used QGroundControl to assist with IMU calibration and verify motor mapping and rotation without propellers.
- Documented mounting, wiring, sensor-detection, and motor-mapping issues in a subsystem checklist and repeated relevant checks after fixes were implemented.

EARLIER DESIGN PROJECTS

Four-Bar Linkage Door-Opening Mechanism

Semester 3

Co-developed and built a four-bar linkage mechanism for door opening as a team project, progressing from a mechanism concept to an assembled linkage and gaining experience with kinematic constraints, link geometry, and mechanism motion.

Wallet for Easier Card Access

Semester 1

Identified limitations of existing card holders in a design-thinking course and developed an alternative wallet concept intended to make card removal easier.

ADDITIONAL PROFESSIONAL EXPERIENCE

Napses Technologies

May 2026 – Jun 2026

Product and Strategy Intern

Hybrid

- Worked with clients, coaches, trainers, nutritionists, and administrators to identify recurring BriskFit problems and translate them into product requirements, interface changes, operating rules, and SOPs.
- Worked with product, design, and operations teams to test onboarding and trainer-client workflow changes; tracked issues through release and post-release review.

Thinkify Labs

May 2025 – Aug 2025

Product and Strategy Intern

Bengaluru, India

- Built an AI-based lead-qualification system using profile and email data to sort LinkedIn leads and route them to appropriate sales representatives, improving operational efficiency by 25%.
- Contributed to a website overhaul and targeted outreach strategy that improved lead conversion by 35% and supported onboarding of seven high-value clients.

ILO Consulting

May 2024 – Jul 2024

Investment Banking Intern

- Built DCF, LBO, and three-statement financial models for M&A due-diligence work on transactions exceeding US\$25M.
- Developed an Ireland market-entry assessment spanning political, regulatory, and competitive risks in three priority sectors and translated the analysis into board-ready recommendations.

LEADERSHIP, MENTORING AND RECOGNITION

Indian Society for Technical Education (ISTE), NITK

Sep 2024 – Present

Secretary

NITK Surathkal

- Led sponsor outreach, budgeting, vendor coordination, and logistics for a technical event attended by 500+ participants; secured three sponsors.

NH66 Fund, P&L Club – Fund Manager

Jan 2025 – Apr 2026

Directed strategy for an INR 150K student-run fund, generating a 10% return over two months and outperforming the benchmark. Mentored three junior analysts in financial modeling, research, and trade execution; organized two recruitment drives and one training workshop.

Global Case Competition at Harvard

2026

Top 50 globally; worked in a four-member team analyzing the European defence landscape and developing structured recommendations.

Campus Venture

NITK

Co-led a three-day campus venture serving 400+ customers with zero stockouts and under 5% wastage.

TECHNICAL SKILLS

Programming

MATLAB, Python, C, C++

Simulation

ANSYS Mechanical; modal and harmonic finite-element analysis; dynamic-system modeling; state estimation; data analysis; regression and machine learning

CAD / Mechanical

SolidWorks; 3D CAD and assembly design; mechanical packaging; mechanism design

Mechatronics / Test

QGroundControl; hardware bring-up; IMU calibration support; motor mapping; system-level troubleshooting; test planning; verification and validation; technical documentation

Languages

English, Hindi, Tamil; French (basic)